# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

The function $L(x) = \int_1^x \frac{dt}{t}$ satisfies the equation

**(A)** $L(x+y) = L(x) + L(y)$

**(B)** $L\left(\frac{x}{y}\right) = L(x) + L(y)$

**(C)** $L(xy) = L(x) + L(y)$

**(D)** none of these

**Q.2**

The value of $\int_0^{2\pi} \max(\sin x, \cos x) dx$ is

**(A)** $3\sqrt{2}$

**(B)** $2\sqrt{2}$

**(C)** 3

**(D)** none of these

**Q.3**

The value of the integral $\int_1^4 e^{\sqrt{x}} dx$

**(A)** $e^2$

**(B)** $2e^2$

**(C)** $4e^2$

**(D)** $3e^2$

**Q.4**

$\lim_{x \to 0} \frac{\int_0^{x^2} \cos t^2 dt}{x \sin x}$ is equal to

**(A)** 1

**(B)** 2

**(C)** 0

**(D)** none of these

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**Q.5**

If $I = \int_0^{\pi/2} \frac{dx}{\sqrt{1+\sin^3 x}}$, then

**(A)** $I \geq \frac{\pi}{2}$

**(B)** $I \geq \frac{\pi}{2\sqrt{2}}$

**(C)** $I \leq \frac{\pi}{2\sqrt{2}}$

**(D)** none of these

**Q.6**

The value of $\int_{-a}^{a} \frac{f(\sin x)}{f(\cos x)+f(\sin^2 x)} dx$, where $f(x)$ is odd function, is

**(A)** cannot be evaluated

**(B)** $I = 0$

**(C)** $I = \pi/2$

**(D)** none of these

**Q.7**

$\frac{1}{c}\int_{ac}^{bc} f\left(\frac{x}{c}\right) dx =$

**(A)** $\frac{1}{c}\int_a^b f(x)dx$

**(B)** $\int_a^b f(x)dx$

**(C)** $c\int_a^b f(x)dx$

**(D)** $\int_{(\mathrm{d})}^{bc^2} f(x)dx$

**Q.8**

If $f(x) = \int_{x^2}^{x^2+1} e^{-t^2} dt$, then $f(x)$ increases in

**(A)** $(2, 2)$

**(B)** no value of $x$

**(C)** $(0, \infty)$

**(D)** $(-\infty, 0)$

**Q.9**

The value of the integral $\int_{-\pi/4}^{\pi/4} \frac{e^x(x \sin x)}{e^{2x}-1} dx$ is e

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**(A)** 0

**(B)** 2

**(C)** e

**(D)** none of these

**Q.10**

$\int_{\alpha}^{\alpha+3\pi} |\sin x| dx$ is equal to

**(A)** 6

**(B)** 3

**(C)** 0

**(D)** none of these

**Q.11**

$\int_{-1}^{1} \cot^{-1}\left(\frac{1}{x}\right) dx$

**(A)** $\pi + \ln 4$

**(B)** $\pi - \ln 2$

**(C)** $\pi + \ln 2$

**(D)** none of these

**Q.12**

Let $f$ be an odd continuous function which is periodic with period $2$. If $g(x) = \int_0^x f(t)dt$

**(A)** $g(x)$ is odd

**(B)** $g(x) = 0, n \in N$

**(C)** $g(2n) = 0, n \in N$

**(D)** $g(x)$ is non-periodic

**Q.13**

Value of $I = \int_0^1 x(1-x)^n dx$ for $n > 2$ is

**(A)** $\frac{1}{n+2}$

**(B)** $\frac{1}{n+1} - \frac{1}{n+2}$

**(C)** $\frac{1}{n+1} + \frac{1}{n+2}$

**(D)** $\frac{1}{n+1}$

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**Q.14**

Let $I_1 = \int_0^3 \frac{\sin x}{\left[\frac{x}{\pi}\right]+\frac{1}{2}}dx$ and $I_2 = \int_{-3}^0 \frac{\sin x}{\left[\frac{x}{\pi}\right]+\frac{1}{2}}dx$ (where $[\cdot]$ is greatest integer function), then

**(A)** $I_1 = 3I_2$

**(B)** $I_2 = 3I_1$

**(C)** $I_1 = I_2$

**(D)** $I_1 + I_2 = 0$

**Q.15**

$\int_0^\pi xf(\sin x)dx$ is equal to

**(A)** $\pi \int_0^\pi f(\sin x)dx$

**(B)** $\frac{\pi}{2} \int_0^{\pi/2} f(\sin x)dx$

**(C)** $\pi \int_0^{\pi/2} f(\cos x)dx$

**(D)** $\pi \int_0^\pi f(\cos x)dx$

**Q.16**

The value of $\int_{-2}^{2} \left(ax^3 + bx + c\right) dx$ depends on

**(A)** the value of $b$

**(B)** the value of $c$

**(C)** the value of $a$

**(D)** the values of $a$ and $b$

**Q.17**

If $\int_a^b |\sin x|dx = 8$ and $\int_0^{a+b} |\cos x|dx = \frac{9}{2}$ and then $a$ is equal to

**(A)** $\frac{\pi}{2}$

**(B)** $\pi$

**(C)** $\frac{\pi}{4}$

**(D)** none of these

**Q.18**

Let $a_n = \int_0^{\pi/4} \tan^n x dx$. Then $a_2 + a_4, a_3 + a_5, a_4 + a_6$ are in

**(A)** A.P.

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**(B)** G.P.

**(C)** H.P.

**(D)** none of these

**Q.19**

If $[x]$ denotes the greatest integer less than or equal to $x$, then the value of $\int_1^5 [|x-3|]dx$ is

**(A)** 1

**(B)** 2

**(C)** 4

**(D)** 8

**Q.20**

If $W_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$ where $n \geq 0$ then which of the following is true?

**(A)** $W_n > W_{n+1}$

**(B)** $W_n < W_{n+1}$

**(C)** $W_3 = 2/5$

**(D)** $W_5 = 8/17$

**Q.21**

Let $I_1 = \int_{1-k}^{k} xf(x(1-x))dx \quad I_2 = \int_{1-k}^{k} f(x(1-x))dx$ where $2k-1 > 0$. Then $\frac{I_2}{I_1}$

**Q.22**

If $\int_{-1}^{1} \frac{t^3+|t|+1}{t^2+2|t|+1}dt = \ln k$. Find the value of $k$.

**Q.23**

If $\int_0^{\pi/4} \frac{\sin 2\theta}{\sin^2\theta+\cos^4\theta}d\theta = \frac{97\sqrt{3}\pi}{A}$, then find the value of $A$.

**Q.24**

$\int_0^{\pi^2/4} \frac{\sin\sqrt{x}}{\sqrt{x}}dx$ is equal to

**Q.25**

The modulus of slope of the normal to the curve $y = \int_0^x \frac{dt}{(1+t^3)}$ at the point where $x = 1$ is

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**Q.26**

If $I=\int_0^1 \cot^{-1}\left(1-x+x^2\right| dx=k \int_0^1 \tan^{-1} x dx$ then $k$ equals

**Q.27**

If $\int_{\sin x}^1 t^2 f(t) dt=1-\sin x, x \in\left(0, \frac{\pi}{2}\right)$, then $f\left(\frac{1}{\sqrt{3}}\right)$ equal to

**Q.28**

The value of $\int_{-2}^0\left[x^3+3 x^2+3 x+3+(x+1) \cos (x+1)\right] dx$ is equal to

**Q.29**

The value of $\lim_{n \to \infty} \frac{1+2^4+3^4+\ldots+n^4}{n^5}-\lim_{n \to \infty} \frac{1+2^3+3^3+\ldots+n^3}{n^5}$ is equal to $1/k$ where $k=$

**Q.30**

$\int_0^{\infty} x^6 e^{-x} dx=$

## Answers & Solutions

**Q.1 Answer:**

$L(xy)=L(x)+L(y)$

**Solution:**

$L(x)=\int_0^x \frac{dt}{t}=\log x$

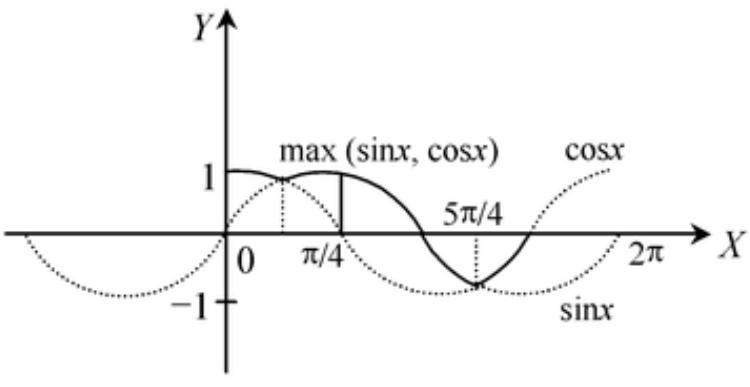
$\therefore \quad L(xy)=\log (xy)=\log x+\log y=L(x)+L(y).$

**Q.2 Answer:**

$2\sqrt{2}$

**Solution:**

$\int_0^{2\pi} \max(\sin x, \cos x) dx=\int_0^{\pi/4} \cos x dx+\int_{\pi/4}^{5\pi/4} \sin x dx+\int_{5\pi/4}^{2\pi} \cos x dx$




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$= |\sin x|_0^{\pi/4} + |-\cos x|_{\pi/4}^{5\pi/4} + |\sin x|_{5\pi/4}^{2\pi} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$

**Q.3 Answer:**

$2e^2$

**Solution:**

$\int_1^4 e^{\sqrt{x}} dx = \int_1^2 e^y 2y dy$ **(put** $x = y^2$**)** $= 2\int_1^2 ye^y dy = 2\left[ye^y - e^y\right]_1^2 = 2e^2$

**Q.4 Answer:**

1

**Solution:**

**Limit=**$\lim_{x\to 0} \frac{\frac{d}{dx}\int_0^{x^2} \cos t^2 dt}{\frac{d}{dx}(x\sin x)} = \lim_{x\to 0} \frac{\cos\left(x^2\right)^2 \cdot \frac{d\left(x^2\right)}{dx}}{\sin x + x\cos x}$

$= \lim_{x\to 0} \frac{2x\cos x^4}{\sin x + x\cos x} = \lim_{x\to 0} \frac{2x\cos x^4}{x\cdot\left(\frac{\sin x}{x}\right) + x\cos x} = \lim_{x\to 0} \frac{2\cos x^4}{1+\cos x} = 1$

**Q.5 Answer:**

$I \geq \frac{\pi}{2\sqrt{2}}$

**Solution:**

**Since** $x \in \left[0, \frac{\pi}{2}\right] \Rightarrow 1 \leq 1 + \sin^3 x \leq 2 \quad \Rightarrow \quad \frac{1}{\sqrt{2}} \leq \frac{1}{\sqrt{1+\sin^3 x}} \leq 1$

$\Rightarrow \quad \int_0^{\pi/2} \frac{1}{\sqrt{2}} dx \leq \int \frac{dx}{\sqrt{1+\sin^3 x}} \leq \int_0^{\pi/2} dx \quad \Rightarrow \quad \frac{\pi}{2\sqrt{2}} \leq I \leq \frac{\pi}{2}$

**Q.6 Answer:**

$I = 0$

**Solution:**

**As** $f(x)$ **is an odd function, the integrand is odd. Hence** $I = 0$

**Q.7 Answer:**

$\int_a^b f(x)dx$

**Solution:**

$I = \frac{1}{c}\int_{ac}^{bc} f\left(\frac{x}{c}\right) dx$

**Put** $z = \frac{x}{c}$**, then** $cdz = dx$

**When** $x = ac, z = a$**; when** $x = bc, z = b$

**Now** $I = \frac{1}{c} \cdot c\int_a^b f(z)dz = \int_a^b f(x)dx$

**Q.8 Answer:**

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$(-\infty, 0)$

**Solution:**

$f'(x) = 2xe^{-\left(x^2+1\right)^2} - 2xe^{-x^4} = 2x\left(e^{-\left(x^2+1\right)^2} - e^{-x^4}\right)$

[Here $e^{-x^4} > e^{-\left(x^2+1\right)^2}$]

So $f'(x) > 0$, when $x < 0$

**Q.9 Answer:**

0

**Solution:**

$I = \int_{-\pi/4}^{\pi/4} \frac{e^x(x \sin x)}{e^{2x}-1} dx$ is zero since integrand is odd.

**Q.10 Answer:**

6

**Solution:**

$\int_{\alpha}^{\alpha+3\pi} |\sin x| dx = 3\int_0^{\pi} \sin x dx = 6$

**Q.11 Answer:**

none of these

**Solution:**

$I = \int_{-1}^{0} \cot^{-1}\left(\frac{1}{x}\right) dx + \int_0^1 \cot^{-1}\left(\frac{1}{x}\right) dx$

$= \int_{-1}^{0} \left(\pi + \tan^{-1} x\right) dx + \int_0^1 \tan^{-1} x dx = \pi + \int_{-1}^{1} \tan^{-1} x dx = \pi$

**Q.12 Answer:**

$g(2n) = 0, n \in N$

**Solution:**

$f$ is an odd function $\Rightarrow \int_0^x f(t)dt$ is an even function

Now $g(x+2) = \int_0^{x+2} f(t)dt = \int_0^2 f(t)dt + \int_2^{x+2} f(t)dt \quad g(2) + \int_0^x f(t)dt$

$= g(2) + g(x)$ ...(i)

$g(2) = \int_0^2 f(t)dt = \int_{-1}^1 f(t+1)dt = 0$ as $f$ is an odd function using in (i)

$\Rightarrow \quad g(x+2) = g(x) \Rightarrow g(x)$ is periodic

$\Rightarrow \quad g(2) = g(4) = g(6) = \ldots\ldots = g(2n) = 0$

**Q.13 Answer:**

$\frac{1}{n+1} - \frac{1}{n+2}$

**Solution:**

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$$I=\int_0^1 x(1-x)^n dx=\int_0^1 (1-x)\cdot x^n dx=\left[\frac{x^{n+1}}{n+1}-\frac{x^{n+2}}{n+2}\right]_0^1=\frac{1}{n+1}-\frac{1}{n+2}$$

**Q.14 Answer:**

$I_1 = I_2$

**Solution:**

$$I_1=\int_0^3 2\sin x dx$$
$$I_2=\int_{-3}^0 \frac{\sin x}{\left[\frac{x}{\pi}\right]+\frac{1}{2}}dx=\int_{-3}^0 \frac{\sin x}{-1+\frac{1}{2}}dx=\int_{-3}^0 2\sin x dx$$
$$=-\int_3^0 2\sin(-x)(-dx)=\int_0^3 2\sin x dx=I_1$$

**Q.15 Answer:**

$\pi\int_0^{\pi/2} f(\cos x)dx$

**Solution:**

$$I=\int_0^\pi xf(\sin x)dx=\int_{=0}^\pi(\pi-x)f(\sin x)dx\Rightarrow 2I=\pi\int_0^\pi f(\sin x)dx$$
$$I=\frac{\pi}{2}\int_0^\pi f(\sin x)dx;\ \textbf{put}\ \frac{\pi}{2}-x=t\quad \textbf{i.e.}\quad I=\frac{\pi}{2}\int_{-\pi/2}^{\pi/2}f(\cos t)dt$$
$$\Rightarrow\quad I=\pi\int_0^{\pi/2}f(\cos x)dx\quad \{\text{as } f(\cos x)\ \textbf{is even}\}$$

**Q.16 Answer:**

the value of $c$

**Solution:**

**Evaluating the given expression we get,**

$$\mathrm{A}=\int_{-2}^{2}\left(\mathrm{ax}^3+\mathrm{bx}+\mathrm{c}\right)\mathrm{dx}$$
$$=\left[\frac{\mathrm{ax}^4}{4}+\frac{\mathrm{bx}^2}{2}+\mathrm{cx}\right]_{-2}^{-2}$$
$$=\left[\frac{\mathrm{a}\left(\left[2^4-(-2)^4\right]\right.}{4}+\frac{\mathrm{b}\left(2^2-(-2)^2\right)}{2}+\mathrm{c}(2-(-2))\right]=0+0+4\mathrm{c}=4\mathrm{c}$$

**So, the final value is only dependent on value of c.**

**Q.17 Answer:**

none of these

**Solution:**

$\int_a^b |\sin x|dx=8\Rightarrow b-a=4\pi$ **as period of** $|\sin x|$ **is** $\pi$ **and** $\int_0^\pi |\sin x|dx=2$

**Also** $|\cos x|$ **is periodic with period** $\pi$ **and** $\int_0^{\pi/2}|\cos x|=1$

**So,** $\int_0^{a+b}|\cos x|dx=\frac{9}{2}\Rightarrow a+b=2\pi+\frac{\pi}{6}\Rightarrow a=-\frac{11\pi}{12}$

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**Q.18 Answer:**

H.P.

**Solution:**

$a_n + a_{n+2} = \int_0^{\pi/4} \left(\tan^n x + \tan^{n+2} x\right) dx, n \geq 2$

$= \int_0^{\pi/4} \tan^n x \sec^2 x dx = \left[\frac{\tan^{n+1} x}{n+1}\right]_0^{\pi/4} = \frac{1}{n+1}$

**Q.19 Answer:**

2

**Solution:**

$I = \int_1^5 [|x-3|]dx \quad \Rightarrow \quad I = \int_1^3 [-1(x-3)]dx + \int_3^5 [(x-3)]dx$

$\Rightarrow \int_1^2 [-(x-3)]dx + \int_2^3 [-(x-3)]dx + \int_3^4 [x-3]dx + \int_4^5 [x-3]dx$

$\Rightarrow I = \int_1^2 dx + \int_2^3 0dx + \int_3^4 0dx + \int_4^5 dx = [x]_1^2 + [x]_4^5$

$\Rightarrow I = I = (2-1) + (5-4) = 2$

**Q.20 Answer:**

$W_n > W_{n+1}$

**Solution:**

**\begin{array}{|c|c|c|c|c|c|c|c|c|c|} \ \hline W_{0} & W_{1} & W_{2} & W_{3} & W_{4} & W_{5} & W_{6} & W_{7} & W_{8} & \ldots & W_{n} \\ \ \hline \frac{\pi}{2} & 1 & \frac{\pi}{4} & \frac{2}{3} & \frac{3 \pi}{16} & \frac{8}{15} & \frac{5 \pi}{32} & \frac{16}{35} & \frac{35 \pi}{256} & \ldots & \frac{n-1}{n} W_{n-2} \\ \ \hline \ \end{array}**

**The sequence $(W_n)$ is decreasing and has positive terms. In fact, for all $n \geq 0$ :**

**$W_n > 0$, because it is an integral of a non-negative continuous function which is not identically zero;**

**-** $W_n - W_{n+1} = \int_0^{\frac{\pi}{2}} \sin^n x dx - \int_0^{\frac{\pi}{2}} \sin^{n+1} x dx = \int_0^{\frac{\pi}{2}} \left(\sin^n x\right)(1 - \sin x) dx > 0$

**Q.21 Answer:**

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**Solution:**

**Given**

$I_1 = \int_{1-k}^{k} xf(x(1-x))dx$ **...(i)**

$I_2 = \int_{1-k}^{k} xf(x(1-x))dx$ **...(ii)**

**Here** $I_1 = \int_{1-k}^{k} xf(x(1-x))dx$

$= \int_{1-k}^{k} (k+1-k-x)f(x+1-k-x)(1-(k+1-k-x))dx$

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$= \int_{1-k}^{k}(1-x)f((1-x)x)dx - \int_{1-k}^{k} xf(x(1-x))dx = \int_{1-k}^{k} f(x(1-x))dx - I_1$
**or** $2I_1 = I_2$ **[using (ii)]**
**or** $\frac{I_1}{I_2} = \frac{1}{2}$

**Q.22 Answer:**

—

**Solution:**

$$I = \int_{-1}^{1} \frac{t^3}{t^2+2|t|+1}dt + \int_{-1}^{1} \frac{(|t|+1)dt}{t^2+2|t|+1}$$

$$0 + 2\int_{0}^{1} \frac{t+1}{t^2+2t+1}dt$$

$= \ln 4 \Rightarrow k = 4 \quad (\because$ first is odd, second is even)

**Q.23 Answer:**

—

**Solution:**

$$\sin^2\theta + cso^4\theta = \frac{1}{2}(1-\cos 2\theta) + \frac{1}{4}(1+\cos 2\theta)^2$$

$$= \frac{1}{4}\left[2 - 2\cos 2\theta + 1 + 2\cos 2\theta + \cos^2 2\theta\right]$$

$$= \frac{1}{4}\left[3 + \cos^2 2\theta\right]$$

$\Rightarrow$ **Given integral** $I = 4\int_0^{\pi/4} \frac{\sin 2\theta}{3+\cos^2 2\theta}d\theta$

**put** $\cos 2\theta = t; -2\sin 2\theta d\theta = dt$

$\Rightarrow \quad I = -2\int_1^0 \frac{dt}{3+t^2} = \frac{2}{\sqrt{3}}\left[\tan^{-1}\left(\frac{t}{\sqrt{3}}\right)\right]_0^1 = \frac{\pi}{3\sqrt{3}} = \frac{97\sqrt{3}\pi}{A}$ given

$\Rightarrow \quad A = 873$

**Q.24 Answer:**

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**Solution:**

**Put** $\sqrt{x} = z, \therefore \quad \frac{1}{2\sqrt{x}}dx = dz$

$\therefore$ given integral $= \int_0^{\pi/2} \sin z.2dz \quad = 2|-\cos z|_0^{\pi/2} = 2\left[-\cos\frac{\pi}{2} + \cos 0\right] = 2$

**Q.25 Answer:**

—

**Solution:**

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$\frac{dy}{dx} = \frac{1}{1+x^3}, \quad \therefore \left.\frac{dy}{dx}\right|_{x=1} = \frac{1}{2}$

**Slope of normal** $m = \frac{-1}{dy/dx}$

**Therefore,** $m = -2$ **and** $|m| = 2$

**Q.26 Answer:**

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**Solution:**

$I = \int_0^1 \tan^{-1}\left(\frac{1}{1-x(1-x)}\right) dx = \int_0^1 \tan^{-1}\left(\frac{(1-x)+x}{1-(1-x)x}\right) dx$

$= \int_0^1 \left[\tan^{-1}(1-x) + \tan^{-1} x\right] dx = \int_0^1 \tan^{-1}(1-x)dx + \int_0^1 \tan^{-1} x dx$

$= \int_0^1 \tan^{-1}(1-(1-x))dx + \int_0^1 \tan^{-1} x dx = 2\int_0^1 \tan^{-1} x dx$

**Q.27 Answer:**

—

**Solution:**

**On differentiating both sides, we get** $-\sin^2 x f(\sin x)\cos x = -\cos x$

$\Rightarrow \quad f(\sin x) = \frac{1}{\sin^2 x} \Rightarrow f(x) = \frac{1}{x^2} \Rightarrow f\left(\frac{1}{\sqrt{3}}\right) = 3$

**Q.28 Answer:**

—

**Solution:**

Let $I = \int_{-2}^0 \left[x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1)dx\right.$

$= \int_{-2}^0 \left[x^3 + 3x^2 + 3x + 1 + 2 + (x+1)\cos(x+1)\right] dx$

$= \int_{-2}^0 \left[(x+1)^3 + 2 + (x+1)\cos(x+1)\right] dx$

**Put** $x + 1 = t \Rightarrow dx = dt$

$\therefore \quad I = \int_{-1}^1 \left(t^3 + 2 + t\cos t\right) dt \quad = \int_{-1}^1 \left(t^3 + t\cos t\right) dt + 2\int_{-1}^1 dt = 0 + 4 \Rightarrow I = 4$

**Q.29 Answer:**

—

**Solution:**

$\lim_{n\to\infty} \frac{1}{n}\left\{\left(\frac{1}{n}\right)^4 + \left(\frac{2}{n}\right)^4 + \left(\frac{3}{n}\right)^4 + \ldots + \left(\frac{n}{n}\right)^4\right\} - \lim_{n\to\infty} \frac{1}{n} \cdot$

$\frac{1}{n}\left\{\left(\frac{1}{n^3}\right) + \left(\frac{2^3}{n^3}\right) + \ldots + \left(\frac{n^3}{n^3}\right)\right\}$

$= \int_0^1 x^4 dx - 0 = \left[\frac{x^5}{5}\right]_0^1 = \frac{1}{5}$

**Q.30 Answer:**

—

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**Solution:**

$I_n = \int_0^\infty t^n e^{-t} dt = [-t^n e^{-t}]_0^\infty + n \int t^{n-1} e^{-t} dt$

$\Rightarrow \quad I_n = nI_{n-1}$ **and** $I_n = 1$

$\Rightarrow \quad I_n = nI_{n-1} = n(n-1)I_{n-2} = n(n-1)(n-2) \ldots \ldots I$

$\Rightarrow \quad I_n = n!$

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